NINE EDUCATION

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[NSG SR-013] JEE MAINS WEEKLY TEST [23-06-2026]

Instructions

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

A. General:

1. This question paper has 22 questions
2. Additional sheets for rough work shall not be provided
3. Blank papers, clipboards, log tables, slide rules, calculators, cellular phones, pagers, and any other electronic gadgets will not be allowed
4. Do not tamper / mutilate the ORS

B. Filling the ORS:

5. Fill in your Roll No., Name and Signature with a pen in appropriate boxes. Do not write these anywhere else
6. Darken the appropriate bubbles below your ID

C. Question paper format and Marking scheme:

7. The question paper is divided into 3 parts; Mathematics , Physics and Chemistry . Each part consists of two sections.
8. In Section A, each question has four choices (1), (2), (3) and (4) out of which ONLY ONE is correct. For every correct choice marked, 4 marks shall be awarded, and for every wrong choice marked 1 mark will be deducted.
9. In Section B, the answer to each question is a numerical value. For every correct answer, 4 marks shall be awarded, and for every wrong answered 1 mark will be deducted.

Mathematics

Physics

Section A

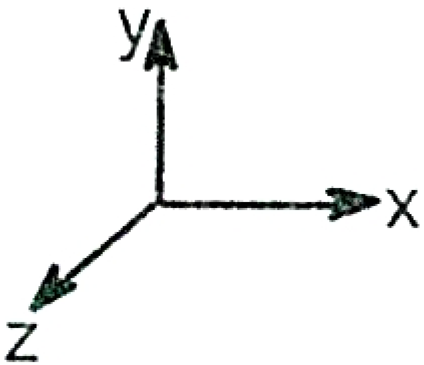
Questions 26-43 contains questions with 4 options (1), (2), (3) and (4), of which only one is correct. Mark the most appropriate choice.

26. (A):A convex lens of glass (μ=1.5) behave as a diverging lens when immersed in carbon disulphide of higher refractive index (μ =1.65).
(R):A diverging lens is thinner in the middle and thicker at the edges.

27. A convex lens of focal length 0.15 m is made of a material of refractive index $\frac{3}{2}$. When it is placed in a liquid, its focal length is increased by 0.225 m. The refractive index of the liquid is

28. The focal length of an equi-convex lens is greater than the radius of curvature of any of the surfaces. Then, the refractive index of the material of the lens is............

29. A coordinate axis as shown below is kept in front of a converging lens at a distance $2f$ from it. Where f is the focal length of the lens. What is the approximate shape of the image as seen by an observer on the other side of the lens at a distance $3f$ from it. Assume that x -axis is the principal axis of the lens:



30. A plano-convex lens fits exactly into a plano-concave lens. Their plane surfaces are parallel to each other. If the lenses are made of different material of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then focal length of the combination is

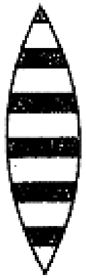
31. An electric charge q exerts a force F on a similar electric charge q separated by a distance r. A third charge $q/4$ is placed midway between the two charges. Now, the force F will

32. It is required to hold equal charges q in equilibrium at the corners of a square. What charge when placed at the center of the square will do this?

33. Electric field on the axis of a small electric dipole at a distance r is $\mathbf{E}_1$ and $\mathbf{E}_2$ at a distance of $2r$ on a line of perpendicular bisector. Then

34. If E_a be the electric field strength of a short dipole at a point on its axial line and E_e that on the equatorial line at the same distance, then

35. A layered lens as shown in the figure is made of two types of transparent materials indicated by different shades. A point object is placed on its axis. The object will form



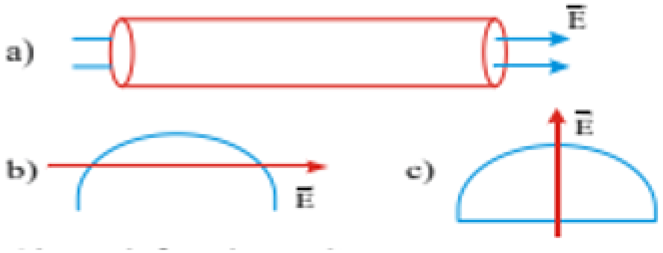
36. How will the image formed by a convex lens be affected, if the central portion of the lens is wrapped in a blank paper, as shown in the figure?

37. A convex lens and a concave lens each having the same focal length of 25 cm are put in a contact with a combination of lenses. The power of the combination is

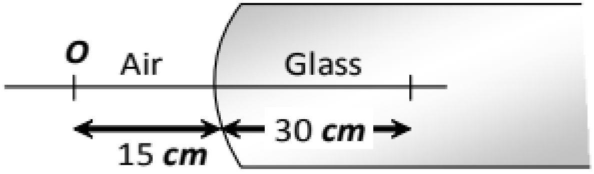
38. A negative charge is placed at the centre of the non-conducting sphere. The direction of the electric field on any point at the surface of the sphere is

39. Electric flux through a surface of area $100\ \mathrm{m}^2$ lying in the xy plane is (in $\mathrm{V-m}$) if $\vec{E} = \hat{i} + \sqrt{2}\hat{j} + \sqrt{3}\hat{k}$

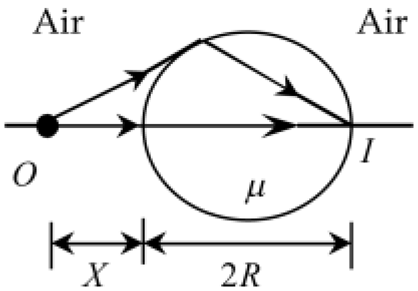
40. In a uniform electric field find the total flux associated with the given surfaces (R is radius),



41. A point object O is placed in front of a glass rod having spherical end of radius of curvature $30\ cm$. the image would be formed at $(\mu = 3/2)$



42. The figure shows a transparent sphere of radius R and refractive index μ. An object O is placed at a distance x from the pole of the first surface, so that a real image is formed at the pole of the exactly opposite surface.



If $x = 2R$, then the value of μ is

43. A uniformly charged and infinitely long line having a linear charge density 'l' is placed at a normal distance y from a point O. Consider a sphere of radius R with O as centre and $\mathrm{R} > \mathrm{y}$. Electric flux through the surface of the sphere is

Section B

Questions 44-47 contains questions of which, the answer would be a numerical value.

44. Consider a uniform charge distribution with charge density of $2\mathrm{C}/\mathrm{m}^3$ throughout in space. If a Gaussian sphere has a variable radius which changes at the rate of $2\mathrm{m}/\mathrm{s}$, then value of rate of change of flux is proportional to r^k, ($r =$ radius of sphere). Then, find the value of k.

45. A fist looking from within water sees the outside world through a circular horizon. If the fish $\sqrt{7}\mathrm{cm}$ below the surface of water. what will be the radius of the circular horizon?(in cm) ($\mu_{water} = 4/3$)

46. When an object is kept at a distance of 30 cm from a concave mirror, the image is formed at a distance of $10\mathrm{cm}$. If the object is moved with a speed of 9 ms^{-1}, the speed with which images moves, is (in ms^{-1})

47. A 16 cm long image of an object is formed by a convex lens on a screen. On moving the lens towards the screen, without changing the position of the object and the screen, 9 cm long image is formed again on the screen. The size of the object is (in cm)

Chemistry